

BIAM: Bilevel Interactive Additive Model for Interpretable Modelling with Missing Values, Noisy Labels and Imbalanced Categories

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Abstract

A triad of imperfections often compromises real-world datasets in high-stakes domains, including weak interpretability, missing values, label noise, and class imbalance, which pose challenges to the deployment of standard machine learning models. This paper introduces the Bilevel Interactive Additive Model (BIAM), a novel framework that concurrently addresses these issues while maintaining interpretability. BIAM’s core innovation lies in its bilevel optimization, which dynamically learns sample weights to counteract the effects of noisy labels and class imbalance. Furthermore, it explicitly models missing values through specific modules, incorporating missingness indicators and their interactions. Furthermore, BIAM leverages piecewise-linear basis functions to capture feature contributions and their interactions, thereby enhancing its extrapolation capabilities in unseen or extreme data regions. Extensive empirical validation demonstrates that BIAM consistently outperforms interpretable and black-box baselines across various data corruption scenarios.

1. Introduction

The proliferation of advanced deep learning models has significantly advanced the field of artificial intelligence and its applications. However, their inherent “black-box” nature raises critical concerns in high-stakes domains such as healthcare, finance, and public safety, where interpretability is as essential as predictive accuracy (Rudin et al., 2022; Agarwal et al., 2021; Duong et al., 2024). Moreover, the performance of these models is highly dependent on the quality of the data. Yet, real-world datasets are often plagued by a triad of imperfections: missing values, label noise, and class imbalance. These issues can severely degrade model perfor-

mance and undermine trust in automated decision systems, making the development of both accurate and interpretable models an urgent priority.

The handling of missing values in supervised learning has been approached through various methodologies. These can be broadly categorized into: (1) impute-then-predict methods, which include single imputation techniques (Shad-bahr et al., 2023) and multiple imputation strategies such as MICE (Van Buuren & Oudshoorn, 1999) and MissForest (Stekhoven & Bühlmann, 2012); (2) models that directly incorporate missingness indicators, as implemented in tree-based methods like XGBoost; and (3) approaches that use surrogate splits (Breiman, 2017) or pattern submodels (Stempfle & Johansson, 2024; McTavish et al., 2024). While these methods offer ways to deal with missingness, they often compromise interpretability or fail to handle label noise and imbalance.

Recently, several studies have explored the use of robust loss functions within interpretable Generalized Additive Models (GAMs) (Yuan et al., 2023; Chen et al., 2021; Wang et al., 2022; Agarwal et al., 2021; Duong et al., 2024) to mitigate the effects of label noise and class imbalance. For instance, Yuan et al. (Yuan et al., 2023) employ a correntropy-induced loss to handle label noise, Chen et al. (2021) utilize ramp loss for enhanced robustness, and Wang et al. (2022) integrate Huber loss for regression tasks under noise. Additionally, the tilted loss functions (Wang et al., 2023) have been incorporated into additive frameworks to address class imbalance. While these approaches improve robustness to specific data imperfections, they require careful manual selection of loss functions and their hyperparameters, which may not generalize across different corruption scenarios. Furthermore, they do not simultaneously handle both label noise and class imbalance, and crucially, they ignore the presence of missing values, which are commonly raised in practical datasets.

This leads to a key challenge:

How to maintain high predictive accuracy and interpretability in the presence of simultaneous missing values, label noise, and class imbalance?

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Table 1. Summary of properties for BIAM and related models ($\checkmark$ = enjoying the property, $\times$ = not available). Notably, "SL", "CE" and "LL" stand for squared loss, cross entropy and Logistic Loss.

Properties	XGBoost	SpAM	HAM	MWNet	GIW	KAN	NAM	NBM	CAT	M-NAM	BIAM
Hypothesis Space	Trees	Splines	RKHS	RKHS	Network	Network	Network	Network	Network	Network	Network
Framework	1-Level	1-Level	1-Level	1-Level	1-Level	1-Level	1-Level	1-Level	1-Level	1-Level	Bilevel
Loss Function	SL / CE	Logistic	Correntropy	Huber	Weighed Loss	SL / CE	SL / CE	SL / CE	SL / CE	LL	Weighed Loss
Regularization	ℓ_1 / ℓ_2	$\ell_{2,1}$	$\ell_{q,1}$	$\ell_{2,1}$	$\times$	$\times$	$\times$	$\times$	$\times$	ℓ_0	$\ell_0 \& \ell_2$
Interpretability *	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Robustness	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Imbalance	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$
Missing Values	$\checkmark$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$

*: Here, we focus on the high-level model-based interpretability (Rudin et al., 2022).

To overcome these limitations, we propose the Bilevel Interactive Additive Model (BIAM). As highlighted in Table 1 compared to the latest baselines, the main contributions of this work are summarized below.

- *Unified Robust Framework*: We introduce a novel bilevel additive model whose lower-level objective jointly handles missing values (via indicators and interactions), label noise, and class imbalance (via meta-learned weights). Its bilevel optimization structure distinctly models missingness patterns while meta-learning a dynamic reweighting scheme to counteract noisy and imbalanced samples.
- *Enhanced Expressivity and Interpretability*: BIAM leverages piecewise-linear basis functions and sparsity-inducing regularization to capture complex feature effects and interactions, yielding superior extrapolation and generalization on unseen data.
- *Comprehensive Empirical Validation*: Through extensive experiments on datasets with various corruptions, BIAM is shown to outperform state-of-the-art interpretable and black-box models consistently.

2. Problem Settings

We consider a supervised learning setting where the training dataset $\mathcal{D}^{tr} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ may contain missing values, noisy labels, and class imbalance. Each feature vector $\mathbf{x}_i \in \mathbb{R}^p$ can have arbitrary missing values, and y_i is a continuous target for regression or a discrete label for classification. For each sample, we define a missingness indicator vector $\mathbf{m}_i \in \{0, 1\}^p$, where $m_i^{(j)} = 1$ if feature j is missing and 0 otherwise. The input includes $\mathbf{x}_i$ and $\mathbf{m}_i$. The data-generating process is complex: labels may be noisy, the class distribution may be imbalanced, and features may be Missing Not at Random (MNAR) or Missing At Random (MAR).

Inspired by M-GAM (McTavish et al., 2024), we model the predictive function as an additive combination of four components: the standard feature effects, the direct effect of missingness, and interaction terms between missingness

indicators and other features. However, unlike M-GAM, which uses piecewise-constant functions, we adopt flexible piecewise-linear hinge functions to improve extrapolation. Furthermore, drawing on the bilevel optimization framework (Shu et al., 2023a; Zhang et al., 2026), we introduce a meta-learning weighting scheme to dynamically assign weights to training samples, thereby enhancing robustness to label noise and class imbalance.

3. Methodology

The architecture of the Bilevel Interactive Additive Model (BIAM) is designed to tackle the outlined challenges through a multi-component framework.

3.1. Additive Prediction Function with Missing Values

To characterize the missing feature values as well as their impacts on other features, the target function of BIAM f , parameterized by $\mathbf{w}$, is formulated as follows

$$f(\mathbf{x}_i, \mathbf{m}_i; \mathbf{w}) = \beta_0 + \underbrace{\sum_{j=1}^p f_j(x_i^{(j)})}_{\text{Standard features}} + \underbrace{\sum_{j=1}^p \beta_j^{miss} m_i^{(j)}}_{\text{Missing indicators}} + \underbrace{\sum_{j=1}^p \sum_{k=1}^p \sum_{\tau=1}^L \alpha_{j,k,\tau} \cdot \mathbb{I}(m_i^{(j)} = 1) \cdot h(x_i^{(k)}; \eta_\tau)}_{\text{Missingness interactions}}$$

where the non-linear effect of each observed feature is modeled by a function f_j . Unlike M-GAM (McTavish et al., 2024) which uses piecewise constants, we adopt piecewise-linear functions from (Yang et al., 2023): $f_j(x_j) = \sum_{\tau=1}^L \beta_j h(x_j; \eta_\tau)$, where $h(x; \eta) = \max(0, x - \eta)$ is a hinge function and η_τ are pre-defined knots. This enables smoother approximations and improved extrapolation. Secondly, the missing indicators β_j^{miss} capture the base effect of the j -th feature being missing. The third counterpart, missingness interactions $(\alpha_{j,k,\tau})$, implies how the shape of another feature k changes when feature j is missing. The interaction is expressed through the same hinge basis $h(x_i^{(k)}; \eta_\tau)$, allowing the adjustment to be non-linear. Furthermore, the threshold set η_τ is obtained following (Yang

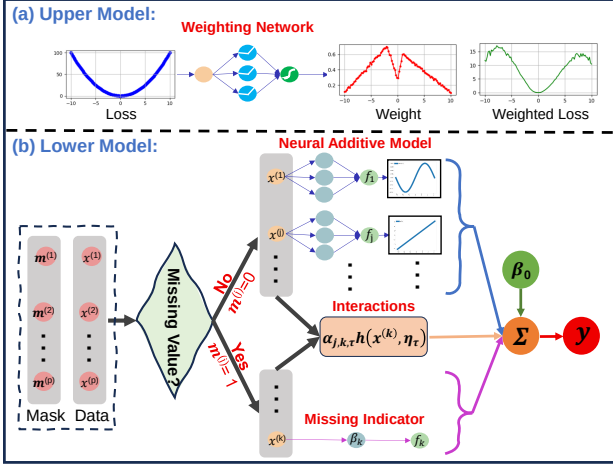


Figure 1. Model architecture of our BIAM, revealing its mechanism of interpretability and robustness against missing values, noisy labels and class imbalance.

et al., 2023). All aforementioned parameters involving f_j , α and β are denoted as $\mathbf{w}$ for simplicity.

3.2. Bilevel Optimization Formulation

The overall architecture of our Bilevel Interactive Additive Model (BIAM) is illustrated in Figure 1, with its mathematical formulation presented below. To jointly address missing values, label noise, and class imbalance, we formulate the learning problem as a bilevel optimization task:

Upper-Level Problem (Validation Loss): The upper-level optimizes the parameters θ of the weighting function ν to minimize the loss on $\mathcal{D}^{val} = \{(\mathbf{x}_i, y_i)\}_{i=n+1}^{n+m}$,

$$\min_{\theta} \mathcal{L} := \frac{1}{m} \sum_{i=n+1}^{n+m} \ell(f(\mathbf{x}_i, \mathbf{m}_i; \hat{\mathbf{w}}(\theta)), y_i), \quad (2)$$

where $\nu(\cdot; \theta)$ is a weighting network (e.g., an MLP) parameterized by θ , learning to assign low weights to samples with high loss like noisy labels, or from the majority class to counterbalance imbalance.

Lower-Level Problem (Weighted Train Loss): The lower-level learns the parameters of the additive model $\mathbf{w} = \{\{\phi_{j,\tau}\}, \{\beta_j^{miss}\}, \{\alpha_{j,k,\tau}\}\}$ on $\mathcal{D}^{tr} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$. The learned weighting function $\nu(\cdot; \theta)$ is equipped in the lower risk to downweight noisy or harmful samples via

$$\hat{\mathbf{w}}(\theta) \in \arg \min_{\mathbf{w}} \mathcal{R}(\theta, \mathbf{m}, \mathbf{w}), \quad (3)$$

where $\mathcal{R}(\theta, \mathbf{m}, \mathbf{w}) := \frac{1}{n} \sum_{i=1}^n \nu(\ell(f(\mathbf{x}_i, \mathbf{m}_i; \mathbf{w}), y_i); \theta) \cdot \ell(f(\mathbf{x}_i, \mathbf{m}_i; \mathbf{w}), y_i) + \lambda_1 \|\mathbf{w}\|_2^2 + \lambda_2 \|\mathbf{w}\|_0$. Here, ℓ is a loss function (e.g., squared loss, logistic loss, or cross-entropy). The ℓ_2 regularization term controls the magnitude of the parameters, while the ℓ_0 norm further promotes sparsity.

These penalties ensure that the most essential features and interactions are retained to preserve interpretability.

3.3. Optimization Algorithm

Solving the bilevel problem directly is computationally challenging. Given initialized $\mathbf{w}^{(0)}$ and $\theta^{(0)}$, we adopt an efficient alternating gradient-based approximation estimator (Ji et al., 2021; Shu et al., 2023b):

Step 1: Lower-Level Step. For fixed θ , compute

$$\mathbf{w}^{(t)} = \mathbf{w}^{(t-1)} - \gamma_w \nabla_{\mathbf{w}} \mathcal{R}(\theta, \mathbf{m}, \mathbf{w}) \Big|_{\theta^{(t)}, \mathbf{w}^{(t-1)}}, \quad (4)$$

and the auxiliary variable

$$\tilde{\mathbf{w}}^{(t)}(\theta) = \mathbf{w}^{(t-1)} - \gamma_w \nabla_{\mathbf{w}} \mathcal{R}(\theta, \mathbf{m}, \mathbf{w}) \Big|_{\theta^{(t)}, \mathbf{w}^{(t)}}. \quad (5)$$

Step 2: Upper-Level Step. Update weighting network parameter θ by differentiating through the fixed inner solution:

$$\theta^{(t+1)} = \theta^{(t)} - \gamma_{\theta} \nabla_{\theta} \mathcal{L}(\tilde{\mathbf{w}}(\theta)) \Big|_{\theta^{(t-1)}, \tilde{\mathbf{w}}^{(t)}}, \quad (6)$$

where $\nabla_{\theta} \mathcal{L}^{val}$ involving the second-order hypergradient is computed via the implicit function theorem or a one-step forward-backward approximation (Shu et al., 2023b).

4. Empirical Validations

This section validates the effectiveness of BIAM on real-world data under various corruptions.

4.1. Experimental Settings

Benchmarks. We validated our model’s performance on Coronal Mass Ejection (Wang et al., 2023) (CME, refer to https://cdaw.gsfc.nasa.gov/CME_list/) and Alzheimer’s Disease Neuroimaging Initiative (ADNI, refer to adni.loni.usc.edu), and two high-dimensional image datasets, MNIST and CelebAMask-HQ (Duong et al., 2024). For the image datasets, we followed the methodology in (Duong et al., 2024) to extract disentangled feature representations before training.

Data Preprocessing. To simulate realistic, imperfect conditions, we systematically introduced three types of corruptions. Let $r_1 \in [0, 1]$ denote the proportion of training samples with noisy labels. For regression, label noise $\epsilon \sim \mathcal{N}(\mu_e, \sigma_e^2)$ is added to the target y . For classification, labels are flipped to an adjacent class. Let r_2 define the imbalance ratio, e.g., a $r_2 = 1 : 10$ ratio means the minority class has one-tenth the samples of the majority class. Finally, let $r_3 \in \{\sqrt{\cdot}, \times\}$ be a missingness indicator of randomly replaced features, injected according to a defined mechanism (e.g., dependent on y or another feature for MAR).

Table 2. Regression performance (average MSE ($\downarrow$) $\pm$ standard deviation) on CME data (upper), and ADNI records (lower). r_1 denotes the ratio of noisy labels, and r_3 indicates the existence of missing values. Top-two results are highlighted in **bold**.

r_1	r_3	MLP	XGBoost	HAM	KAN	NAM	CAT	BIAM ^{-B}	BIAM ^{-I}	BIAM ^{-H}	BIAM
0	$\times$	0.598 $\pm$ 0.129	0.604 $\pm$ 0.140	0.626 $\pm$ 0.144	0.608 $\pm$ 0.103	0.619 $\pm$ 0.116	0.585 $\pm$ 0.107	0.593 $\pm$ 0.113	0.581 $\pm$ 0.108	0.575 $\pm$ 0.099	0.570 $\pm$ 0.081
10%	$\times$	0.863 $\pm$ 0.217	0.844 $\pm$ 0.194	0.705 $\pm$ 0.173	0.858 $\pm$ 0.186	0.874 $\pm$ 0.199	0.830 $\pm$ 0.155	0.837 $\pm$ 0.149	0.621 $\pm$ 0.134	0.612 $\pm$ 0.121	0.603 $\pm$ 0.114
30%	$\times$	1.611 $\pm$ 0.743	1.581 $\pm$ 0.726	0.740 $\pm$ 0.183	1.577 $\pm$ 0.512	1.603 $\pm$ 0.481	1.310 $\pm$ 0.422	1.317 $\pm$ 0.430	0.669 $\pm$ 0.162	0.671 $\pm$ 0.158	0.644 $\pm$ 0.147
0	$\checkmark$	0.616 $\pm$ 0.142	0.612 $\pm$ 0.141	0.632 $\pm$ 0.154	0.626 $\pm$ 0.107	0.635 $\pm$ 0.128	0.596 $\pm$ 0.119	0.613 $\pm$ 0.125	0.585 $\pm$ 0.117	0.588 $\pm$ 0.116	0.581 $\pm$ 0.094
10%	$\checkmark$	0.891 $\pm$ 0.232	0.862 $\pm$ 0.206	0.721 $\pm$ 0.184	0.886 $\pm$ 0.207	0.902 $\pm$ 0.219	0.857 $\pm$ 0.165	0.865 $\pm$ 0.163	0.646 $\pm$ 0.151	0.638 $\pm$ 0.139	0.633 $\pm$ 0.130
30%	$\checkmark$	1.637 $\pm$ 0.761	1.606 $\pm$ 0.740	0.767 $\pm$ 0.195	1.603 $\pm$ 0.521	1.631 $\pm$ 0.496	1.337 $\pm$ 0.436	1.345 $\pm$ 0.441	0.697 $\pm$ 0.174	0.696 $\pm$ 0.178	0.672 $\pm$ 0.162
0	$\times$	0.174 $\pm$ 0.042	0.167 $\pm$ 0.037	0.168 $\pm$ 0.040	0.184 $\pm$ 0.051	0.190 $\pm$ 0.048	0.175 $\pm$ 0.045	0.181 $\pm$ 0.047	0.173 $\pm$ 0.056	0.174 $\pm$ 0.052	0.172 $\pm$ 0.045
10%	$\times$	0.342 $\pm$ 0.091	0.336 $\pm$ 0.094	0.192 $\pm$ 0.059	0.361 $\pm$ 0.120	0.372 $\pm$ 0.114	0.324 $\pm$ 0.092	0.326 $\pm$ 0.094	0.192 $\pm$ 0.077	0.191 $\pm$ 0.076	0.183 $\pm$ 0.068
30%	$\times$	0.746 $\pm$ 0.139	0.739 $\pm$ 0.128	0.251 $\pm$ 0.088	0.701 $\pm$ 0.122	0.713 $\pm$ 0.132	0.649 $\pm$ 0.102	0.660 $\pm$ 0.105	0.235 $\pm$ 0.098	0.233 $\pm$ 0.097	0.225 $\pm$ 0.091
0	$\checkmark$	0.182 $\pm$ 0.053	0.176 $\pm$ 0.047	0.176 $\pm$ 0.046	0.192 $\pm$ 0.065	0.197 $\pm$ 0.056	0.183 $\pm$ 0.053	0.189 $\pm$ 0.058	0.181 $\pm$ 0.069	0.184 $\pm$ 0.066	0.179 $\pm$ 0.057
10%	$\checkmark$	0.352 $\pm$ 0.107	0.344 $\pm$ 0.108	0.203 $\pm$ 0.066	0.371 $\pm$ 0.130	0.381 $\pm$ 0.122	0.331 $\pm$ 0.108	0.335 $\pm$ 0.108	0.211 $\pm$ 0.087	0.204 $\pm$ 0.086	0.191 $\pm$ 0.076
30%	$\checkmark$	0.757 $\pm$ 0.149	0.745 $\pm$ 0.134	0.261 $\pm$ 0.095	0.711 $\pm$ 0.134	0.722 $\pm$ 0.140	0.657 $\pm$ 0.109	0.667 $\pm$ 0.112	0.245 $\pm$ 0.108	0.241 $\pm$ 0.107	0.235 $\pm$ 0.098

Table 3. Classification performance (Macro-F1 score $\uparrow$ $\pm$ standard deviation) evaluated on MNIST (upper) and CelebAMask-HQ (lower) datasets. Notably, r_1 , r_2 , and r_3 represent the noise ratio, class imbalance ratio, and the value missingness.

r_1	r_2	r_3	MLP	XGBoost	CSAM	RSAM	GIW	KAN	NAM	CAT	M-NAM	BIAM
0	1:1	$\times$	0.942 $\pm$ 0.031	0.941 $\pm$ 0.040	0.921 $\pm$ 0.057	0.917 $\pm$ 0.077	0.933 $\pm$ 0.034	0.901 $\pm$ 0.055	0.864 $\pm$ 0.046	0.949 $\pm$ 0.031	0.942 $\pm$ 0.052	0.951 $\pm$ 0.033
0	1:10	$\times$	0.891 $\pm$ 0.068	0.894 $\pm$ 0.067	0.864 $\pm$ 0.071	0.860 $\pm$ 0.083	0.912 $\pm$ 0.051	0.853 $\pm$ 0.096	0.831 $\pm$ 0.109	0.898 $\pm$ 0.053	0.894 $\pm$ 0.063	0.927 $\pm$ 0.050
20%	1:10	$\times$	0.830 $\pm$ 0.124	0.834 $\pm$ 0.117	0.849 $\pm$ 0.088	0.851 $\pm$ 0.091	0.903 $\pm$ 0.074	0.820 $\pm$ 0.110	0.804 $\pm$ 0.166	0.839 $\pm$ 0.117	0.864 $\pm$ 0.089	0.916 $\pm$ 0.073
0	1:1	$\checkmark$	0.923 $\pm$ 0.044	0.934 $\pm$ 0.041	0.903 $\pm$ 0.065	0.896 $\pm$ 0.086	0.912 $\pm$ 0.047	0.891 $\pm$ 0.065	0.845 $\pm$ 0.056	0.931 $\pm$ 0.046	0.933 $\pm$ 0.062	0.943 $\pm$ 0.041
0	1:10	$\checkmark$	0.872 $\pm$ 0.078	0.882 $\pm$ 0.065	0.846 $\pm$ 0.080	0.842 $\pm$ 0.098	0.902 $\pm$ 0.066	0.833 $\pm$ 0.108	0.814 $\pm$ 0.116	0.887 $\pm$ 0.069	0.874 $\pm$ 0.077	0.917 $\pm$ 0.065
20%	1:10	$\checkmark$	0.813 $\pm$ 0.132	0.823 $\pm$ 0.115	0.832 $\pm$ 0.095	0.830 $\pm$ 0.101	0.893 $\pm$ 0.084	0.802 $\pm$ 0.128	0.783 $\pm$ 0.171	0.821 $\pm$ 0.126	0.844 $\pm$ 0.107	0.906 $\pm$ 0.081
0	1:1	$\times$	0.766 $\pm$ 0.071	0.754 $\pm$ 0.076	0.721 $\pm$ 0.082	0.723 $\pm$ 0.082	0.730 $\pm$ 0.101	0.746 $\pm$ 0.098	0.723 $\pm$ 0.113	0.740 $\pm$ 0.087	0.757 $\pm$ 0.090	0.773 $\pm$ 0.065
0	1:10	$\times$	0.673 $\pm$ 0.106	0.661 $\pm$ 0.114	0.637 $\pm$ 0.122	0.644 $\pm$ 0.110	0.725 $\pm$ 0.127	0.671 $\pm$ 0.124	0.630 $\pm$ 0.135	0.651 $\pm$ 0.104	0.650 $\pm$ 0.105	0.754 $\pm$ 0.073
20%	1:10	$\times$	0.611 $\pm$ 0.137	0.594 $\pm$ 0.142	0.620 $\pm$ 0.133	0.617 $\pm$ 0.130	0.712 $\pm$ 0.142	0.614 $\pm$ 0.148	0.599 $\pm$ 0.157	0.618 $\pm$ 0.121	0.625 $\pm$ 0.116	0.740 $\pm$ 0.088
0	1:1	$\checkmark$	0.736 $\pm$ 0.087	0.743 $\pm$ 0.086	0.702 $\pm$ 0.095	0.704 $\pm$ 0.098	0.711 $\pm$ 0.106	0.727 $\pm$ 0.095	0.704 $\pm$ 0.116	0.721 $\pm$ 0.081	0.746 $\pm$ 0.090	0.762 $\pm$ 0.077
0	1:10	$\checkmark$	0.651 $\pm$ 0.132	0.658 $\pm$ 0.115	0.626 $\pm$ 0.122	0.632 $\pm$ 0.118	0.714 $\pm$ 0.133	0.657 $\pm$ 0.125	0.616 $\pm$ 0.132	0.637 $\pm$ 0.108	0.647 $\pm$ 0.106	0.742 $\pm$ 0.081
20%	1:10	$\checkmark$	0.587 $\pm$ 0.141	0.585 $\pm$ 0.143	0.607 $\pm$ 0.133	0.603 $\pm$ 0.138	0.703 $\pm$ 0.149	0.604 $\pm$ 0.155	0.586 $\pm$ 0.161	0.608 $\pm$ 0.120	0.622 $\pm$ 0.118	0.725 $\pm$ 0.092

Baselines and Metrics. We compare BIAM against 10 state-of-the-art methods spanning five categories: (1) standard models (MLP, XGBoost); (2) robust additive models (CSAM (Yuan et al., 2023), RSAM (Chen et al., 2021), HAM (Wang et al., 2022)); (3) imbalance-handling approaches (GIW (Fang et al., 2024)); (4) interpretable neural architectures (NAM (Agarwal et al., 2021), KAN (Liu et al., 2025), M-NAM (McTavish et al., 2024)); and (5) concept-based models (CAT (Duong et al., 2024)). This comprehensive selection enables systematic evaluation across interpretability, robustness, and class-imbalance handling. Notably, the median-based imputation strategy is used for these baselines before training, except for XGBoost and M-GAM. We evaluated regression performance using the Mean Squared Error (MSE) and classification performance using the Macro-F1 score, as the latter criterion is more suitable for imbalanced datasets.

4.2. Results and Analysis

As presented in Table 2, BIAM consistently provides competitive and superior performance compared to baselines, especially under conditions of severe data corruption. For example, on the ADNI dataset with 30% label noise and the presence of missing values, BIAM achieves an MSE of 0.235. This result marks a significant improvement over the latest NAM-based baselines, M-NAM and CAT, and is substantially better than standard XGBoost.

Our ablation studies in Table 2 confirm the necessity of BIAM’s core components. Removing the bilevel optimization

(BIAM^{-B}) causes a sharp drop in F1 score due to noise and imbalance. Removing the missingness interactions (BIAM^{-I}) hurts performance when $r_3=\checkmark$. Replacing piecewise-linear hinges with piecewise constants (BIAM^{-H}) results in worse extrapolation and feature interactions, particularly with missing values $r_3 = \checkmark$.

In the more complex classification tasks (Table 3), BIAM’s robustness is even more pronounced. On the CelebAMask-HQ dataset, under the most challenging scenario combining $r_1=20\%$ label noise, the imbalanced ratio of $r_2=1:10$, and missing values $r_3=\checkmark$, BIAM attains a Macro-F1 score of 0.725 ± 0.092 . This represents a remarkable performance gain of over 10 absolute percentage points compared to M-NAM (McTavish et al., 2024), which can also handle missing values. The widening performance gap between BIAM and other models, as data quality degrades, highlights the effectiveness of its unified architecture in practice.

5. Conclusion

In this work, we introduced BIAM, an interpretable framework designed to robustly handle the simultaneous presence of missing values, noisy labels, and class imbalance. The novel model structure integrates an adaptive sample reweighting scheme to counteract noise and imbalance, along with an explicit, sparse model for missingness patterns and their interactions, thereby achieving both strong robustness and interpretability in critical real-world applications.

Impact Statement

This paper presents work aimed at advancing the foundations of interpretable machine learning and robust modeling under complex, imperfect data conditions. We believe this work can deepen our understanding of adaptive sample reweighting and complex feature interactions within a bilevel optimization framework, and provide trustworthy modeling tools for automated decision-making in high-stakes domains such as healthcare, finance, and public safety.

The proposed unified framework significantly bridges the critical gap between strong empirical robustness and model transparency, compared with the existing literature, which is primarily focused on addressing data imperfections, such as missing values, label noise, or class imbalance, in isolation or relies heavily on black-box architectures. There may be some potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Appendix

A. Limitations and Discussions

The comprehensive empirical robustness achieved by the BIAM framework naturally introduces specific computational and structural trade-offs. Formulating the objective as a hierarchical optimization problem necessitates alternating gradient updates and hypergradient approximations, which inherently demand greater memory capacity and processing time compared to standard single-level estimators. Moreover, the explicit modeling of pairwise missingness interactions causes the parameter space to expand quadratically with the feature dimension. Consequently, deploying this architecture in extremely high-dimensional environments requires strict enforcement of sparse regularization to maintain computational feasibility without sacrificing the intended model transparency.

A secondary constraint lies in the current feature representation mechanism, which relies on predefined knots for the piecewise-linear basis functions. While this design provides smoother approximations and better extrapolation than piecewise-constant alternatives, fixed knot placement may restrict the model’s capacity to autonomously capture highly localized or complex nonlinear effects. Future investigations will prioritize adaptive knot selection strategies and explore higher-order missingness dependencies to further enhance structural flexibility. Nevertheless, these initialization and scaling constraints do not undermine the methodological validity of the approach. By successfully balancing rigorous empirical validation with strict interpretability under severe data corruption, this work establishes a highly effective foundation for trustworthy decision-making in complex, real-world applications.

B. Ethics Statements

This research does not raise any ethical concerns. The authors confirm that this work was conducted in accordance with the principles of academic integrity and research ethics.

C. Reproducibility Statement

We ensure full reproducibility by publicly releasing relevant materials, code, and data resources. All experiments are implemented in Python on a PC equipped with an Intel(R) Xeon(R) Platinum 8175M CPU and an NVIDIA RTX A6000 GPU. The implementations are available at <https://anonymous.4open.science/r/BIAM-6772>.

Related results were collected using fixed computational resources detailed in Section 4. This enables independent verification of the research’s findings.

D. Disclosure of Generative AI Usage

GenAI tools were used during the editing (e.g., grammar, spelling, word choice). And the authors are fully accountable for the content.